

20 — Level 1 rejects DP: the installed engines cannot carry the demand

Proves · Level 1 rejecting a mode because the installed engines cannot carry its demand.

Mechanics this scenario turns on

- Every active mode runs its **own** Level 1 — own demand, own combinations, own baseline, own optimum, own t/h. There is no single "tonnes per hour" for the vessel; the year is $\sum (\text{mode t/h} \times \text{mode hours})$.
- Validation runs **before** Level 1. A plant that cannot carry its load is rejected there, with a different message and a different code path from the Level 1 rejections.

Panels described below · Why the failure has to be in DP, not Transit · The response

Anything not described here — the mechanics above name the step that produced it; **00-ORIENTATION** Part 6 has the full number-to-step index.

Trust · characterisation snapshot, generated from the code. It detects change; it does NOT prove correctness. Figures marked *pending reference verification* have never been checked against anything outside the application.

Read after · scenario 17.

One of three scenarios (20, 21, 22) added because **only 2 of the 4 rejection messages** a user can receive had end-to-end coverage. **Level1RejectionTally.ExplainFor** picks one sentence based on which counter fired, and that sentence is the entire actionable content of the 400.

Why the failure has to be in DP, not Transit

ValidationService checks main-engine utilisation **for Transit only**. Any Transit plant short of power is rejected there, before Level 1 runs — that path is scenario 17. To reach Level 1's *own* rejection the shortfall must be in a mode validation does not inspect.

Transit : propulsion 4 000, hotel 800	→ ME utilisation 40 %	✓ passes validation
DP : required thrust 10 200	→ ME needs > 10 000 kW	✗ no combination survives

The plant is $2 \times 5\,000$ kW main engines with no shaft generator, so 10 200 kW of thrust cannot be produced by any on/off combination, and PTI is not configured.

The response

400

No feasible engine configuration: the installed engines cannot carry the DP demand.

Increase engine capacity or engine count, or reduce the propulsion/hotel power for this mode.

Two details worth noticing:

- **The mode is named.** A user with four modes configured needs to know which one failed.
- **The advice is about capacity,** not about the battery or PTI — those branches take precedence in `ExplainFor` and did not fire here.

Takeaway: validation guards Transit; Level 1 guards every other mode. A vessel can pass every form check and still be impossible to operate in DP.